Nathan R.T. Lesnevich

Personal Information

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Academic Positions

2024-current Postdoctoral Fellow, Oklahoma State University

Education

May 2024 Ph.D. in Mathematics, **Washington University in St. Louis** (WashU), St. Louis MO
Thesis: Splines on Cayley Graphs of the Symmetric Group.
| Advisors: Assoc. Prof. Martha Precup
| Prof. John Shareshian

May 2021 A.M. in Mathematics, **Washington University in St. Louis** (WashU), St. Louis MO

May 2019 B.S. in Mathematics (*magna cum laude*), **Bucknell University**, Lewisburg PA
Pure Mathematics concentration, additional fulfillment of B.A. in Computer Science.
Thesis: Positivity Among P-Partition Generating Functions
| Advisor: Prof. Peter McNamara

Preprints and Publications

9. J. Boretsky, R. Hodges, N. Lesnevich, T. McAllister, and J. Pappe. *Dot actions and Hessenberg spaces for dihedral groups*. (2026) [arXiv:2610.16001]
8. A. Dave, T. Feng, N. Lesnevich, S. Oh, E. Richmond, E. Wang, and A. Wu. *Shifted lower Bruhat intervals are EL-shellable*. submitted (2026). [arXiv:2608.04417]
7. J. Boretsky, A. Cornejo, R. Hodges, P. Horn, N. Lesnevich, and T. McAllister. *Comparability in Bruhat orders*. submitted (2026). [arXiv:2601.21158]
6. N.R.T. Lesnevich. *Signed permutations and degree-one dot action representations for types B and C*. submitted (2025). [arXiv:2511.13913]
5. J. Lee, N. Lesnevich, and M. Precup. *Restricted inversion polynomials*. to appear in Electronic Journal of Combinatorics (2025). [arXiv:2511.05676]
4. B. Adenbaum, E. Barnard, M. Hlavacek, B. Kagy, N. Lesnevich, G. D. Nasr, and K. Waddle. *The poset of maximal tubings of the cycle graph is a lattice*. submitted (2025). [arXiv:2510.09429]
3. N.R.T. Lesnevich. *Splines on Cayley graphs of the symmetric group*. Forum of Mathematics, Sigma 13 (2025), e96. [arXiv:2401.14380]
2. N.R.T. Lesnevich. *Hook-shape immanant characters from Stanley–Stembridge characters*. Algebraic Combinatorics 7 (2024), no. 1, 137-157. [arXiv:2304.05285]
1. N.R.T. Lesnevich and P.R.W. McNamara. *Positivity among P-partition generating functions*. Annals of Combinatorics 26 (2022), 171-204. [arXiv:2006.10087]

Grants and Funding

2023 | Prepared and received NSF conference grant DMS-2245927 (PI: Martha Precup)
Graduate Student Combinatorics Conference 2023
Co-written with a small group of other graduate students

Teaching

2024-Current | Instructor, Oklahoma State University
Two sections per semester, instructor of record.
–Graduate Algebra (FL26)
–Undergraduate Combinatorics (SP26)
–Calculus II (SP26)
–Differential Equations (FL24, SP25, FL25, FL26)

2020-2023 | Graduate Teaching Assistant, Washington University in St. Louis
–Finite Math (SU22, instructor of record)
–Calculus II (SP21)
–Calculus III (FL20, SP23)
–Differential Equations (FL21, SP22, FL22)

2022-2024 | Organizer, Washington University Graduate Directed Reading Program

2021-2023 | Instructor, Washington University Graduate Directed Reading Program
–Representation Theory (FL23)
–Lie Algebras (SP22)
–Ehrhart Theory (FL21)
–Young Tableaux (SP21)

Awards | Robert H. McDowell Teaching Award (2023).
Department graduate teaching award, 1 annual awardee.
Brian Blank Award for third-year students (2022).
Department graduate award, 1-2 annual awardees.

Programs and Outreach

2026 | Speaker, OSU High School Math Day, Oklahoma State University
Error Correcting Codes

2024 | Research advisor, Freiwald Scholars Summer Research Program, Washington University in St. Louis
Co-advised several undergraduate research projects

2024 | Assistant, Sonia Kovalevsky Day at Washington University in St. Louis
Cryptography Activity Organizer

2022-2024 | Organizer, Graduate Directed Reading Program, Washington University in St. Louis
Recruitment, Organizational, and Administrative Duties

2023 | Speaker, Math Circle, Washington University in St. Louis
Cryptography: The math of making and breaking codes

Conferences and Seminars

Organization

Spring 2025	Special Session on Combinatorics and Lie Theory AMS Spring Central Sectional Meeting *University of Kansas*
Spring 2023	Graduate Student Combinatorics Conference 2023 Co-wrote and received NSF conference grant DMS-2245927.
2022-2023	WashU Combinatorics Seminar

Speaking

Fall 2026	Smith Colloquium, University of Kansas *Splines, Graphs, and Symmetry*, Sept 17
Fall 2026	KU Combinatorics Seminar *Shifted Lower Bruhat Intervals are EL-Shellable*, Sept 18
Spring 2026	The CAGE Seminar (inter-university Philadelphia) *GKM Theory and Reflection Groups*, April 16
Spring 2026	AMS Spring Southeastern Sectional, Special Session on Symmetric functions and subvarieties of flag varieties *Some type B/C symmetric functions from Hessenberg varieties*, March 29
Spring 2026	AMS Spring Southeastern Sectional, Special Session on Algebraic Combinatorics *The maximal tubings of the cycle graph form a lattice*, March 28
Fall 2025	The Schubert Seminar (inter-university) *Combinatorics for GKM Theory and the Dot Action*, Nov 3
Fall 2025	AMS Fall Central Sectional, Special Session on Interactions between Geometry, Combinatorics, and Flag Varieties *The Second Cohomology of Hess(X,H) and the Dot Action in types B/C*, Oct 19
Summer 2025	Graduate Research Workshop in Combinatorics (at Iowa State) *Dihedral Splines*, June 15-27
Spring 2025	WashU Combinatorics Seminar *The Maximal Tubings of the Cycle Graph and G-inversions*, March 17
Spring 2024	Poster, Enumerative and Algebraic Combinatorics (at UFL) *Splines on Cayley Graphs of the Symmetric Group*, Feb 25-27
Fall 2023	KU Combinatorics Seminar *Splines on Cayley Graphs of the Symmetric Group*, Nov 3
Summer 2023	Short Talk, Schubert Conference (at UIUC) *Splines on Cayley Graphs of the Symmetric Group*, June 20
Spring 2023	Poster, AlGeCom-XXIII *Splines on Cayley Graphs of the Symmetric Group*, Apr 15
Spring 2023	Graduate Online Combinatorics Colloquium *Splines on Cayley Graphs of the Symmetric Group*, Mar 8
Spring 2023	Dartmouth Combinatorics Seminar *Splines on Cayley Graphs of the Symmetric Group*, Feb 21

Fall 2022	AMS Fall Eastern Sectional, Special Session on Geometric Aspects of Algebraic Combinatorics. *A Proof of an Immanant Conjecture for Hook Partitions*
Spring 2022	Graduate Student Combinatorics Conference 2022 *Hook-Partition Immanant Characters from Stanley–Stembridge Characters*

Recent Program/Workshop Attendance

Summer 2026	CMND 2026 Thematic Program: Algebraic Combinatorics and Applications, at University of Notre Dame
Summer 2025	Graduate Research Workshop in Combinatorics, at Iowa State University
Summer 2024	AMS Mathematics Research Communities: Algebraic Combinatorics, at Beaver Hollow Convention Center
Spring 2024	BIRS Workshop: Homological Perspectives on Splines, at UBC Kelowna
Summer 2023	Schubert Calculus Summer School, at UIUC
Spring 2023	8th Lake Michigan Workshop on Combinatorics and Graph Theory, at the University of Notre Dame
Spring 2022	ISM Discovery School: Geometry and Combinatorics of Hessenberg Varieties, at UQAM.

Scholarships and Honors

2020-2021	David B. Emmes Endowed Scholarship.
2015-2019	John C. Hoover Scholarship for Students in Mathematics
2015-2019	Bucknell Presidential Fellowship *4-year half-tuition merit scholarship with research mentorship.*
2015-2019	Bucknell University's Dean's List
2018	John C. Hoover Summer Research Funding Recipient
2017	Pi Mu Epsilon Mathematics Honors Society Member
2017	Bucknell Program for Undergraduate Research Award Recipient